

CER-AI PREOPERATIVE ECTASIA RISK ASSESSMENT

Corneal Ectasia Risk Assessment Intelligence

The final surgical decision and all associated responsibility and liability rest with the surgeon. This application is a clinical decision-support aid only.

Patient	Sample 01	Age	27
Assessment date	Removed	Reviewer	Removed
Eyes	OD; OS	Overall disposition	PASS

PASS — final CER-AI combination criteria met; review individual findings.

OD — PASS

Randleman / ERSS

Component	Points / disposition	Exact finding
Topography	0	
Residual stromal bed	0	
Age	0	
Preoperative pachymetry	0	
Manifest MRSE	0	
Total	0	
Topography category	NORMAL_SYMMETRIC	
Disposition	PASS	

NICE

Component	Input	Points
K2	42.7	1
central_pachymetry	517	2
B_Ele_Th	4	1
I_S	0.03	1
Total		5
Classification	CAUTION	CAUTION
Decision trigger	CAUTION	NICE total: 5

PS3

Factor	Status	Exact finding
anterior_km	NORMAL	Anterior Km 41.9 D < 48 D.
thinnest	NORMAL	Thinnest 516 µm >=500 µm.
elevation	NORMAL	F.Ele.Th 3 µm <=12 and B.Ele.Th 4 µm <=15.
inter_eye_asymmetry	NORMAL	Inter-eye score 1/5; exceeded: anterior Km.
srax	NORMAL	Front-map SRAX 13.5° <= 20°. Source: Axial/Sagittal Curvature (Front).
ppi_average	NORMAL	PPI Average 1.09 <= 1.20.
corneal_thickness_map_morphology	NOT_EVALUATED	Corneal Thickness Map morphology: not evaluated; surgeon review required; not counted in automated PS3.
relative_thickness_map	NOT_EVALUATED	Relative Thickness Map: not evaluated; surgeon review required; not counted in automated PS3.
pti_ctsp_morphology	NOT_EVALUATED	PTI/CTSP thickness-profile morphology: not evaluated; surgeon review required; not counted in automated PS3.
Moderate / High	0	0
Procedure disposition	PASS	Raw PS3 PRK disposition: ALLOWED.

Astigmatic disparity validation

Status	Magnitude difference	Axis difference
VALIDATION_REQUIRED	1.1	55
Interpretation	Astigmatic disparity (magnitude difference 1.10 D; axis difference 55.0°) requires measurement/refraction validation; it is not a PS3 risk factor and does not independently defer LASIK.	Non-scoring; no independent procedure restriction

Belin/Ambrósio BAD-D

Parameter	Value	Interpretation
Final BAD-D	0.51	NORMAL / PASS
Df	0.33	NORMAL / < 1.60; information only
Db	-0.41	NORMAL / < 1.60; information only
Dp	1.26	NORMAL / < 1.60; information only
Dt	0.65	NORMAL / < 1.60; information only
Da	0.83	NORMAL / < 1.60; information only
PPI Min	0.8	SUSPICIOUS / 0.80-0.86; information only
PPI Avg	1.09	SUSPICIOUS / 1.08-1.17; information only
PPI Max	1.5	SUSPICIOUS / 1.40-1.52; information only
ARTmax	397	NORMAL / > 368 µm; information only

Procedural safety

Parameter	Canonical result	Exact finding
LASIK_RSB_um	Not documented	
PRK_RST_um	417.25	
LASIK_PTA_percent	Not documented	
PRK_PTA_percent	19.138	
estimated_final_Kmean_D	39.3	
scalar_final_Kmean_mode l_valid	Yes	
final_Kmean_model	MYOPIC_0.8_D_PER_ D_ESTIMATE	
hyperopic_review_level	Not documented	
hyperopic_treatment_com ponent_D	-3	
hard_stops	preop_thickness: No; sphere_magnitude: No; lasik_rsb: No; lasik_pta: No; prk_rst: No; prk_pta: No; final_kmean: No	
missing	Not documented	
status	PASS	

Procedure planning

Planning item	Canonical result
OD — PASS — PRK	OZ 6.5 mm — TZ 9 mm — Ablation 48.75 µm
Selected LASIK plan	Not documented
sequence	Not documented
rejection_reasons	Not documented
MMC guidance	RECOMMENDED

Decision basis

Level	Canonical decision driver
CAUTION	key: nice; status: CAUTION; detail: NICE total: 5

Canonical Pentacam values and provenance

Canonical field	Value	Provenance
K1	41.1	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT /
K2	42.7	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT /
K2 axis	102.5	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT /
Km	41.9	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT /
Astigmatism	1.6	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT /
Displayed steep/astigmatic axis	102.5	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT /
PS3 BAD Axis (flat meridian, beside K1)	125	BELIN_AMBROSIO_CENTER_NUMERIC_BOX /
ISV	17	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM /
IVA	0.09	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM /
KI	1.01	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM /
CKI	1	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM /
IHA	0.1	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM /
IHD	0.008	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM /
Topometric RMin	7.87	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM /
KISA	5.059	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM /
Signed I-S	0.03	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM /
Pupil Center pachymetry	517	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX /
Thinnest pachymetry	516	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX /
Kmax (Front)	42.9	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX /
HWTW	12.1	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX /
F.Ele.Th	3	BELIN_AMBROSIO_LOWER_ELEVATION_LABEL_VALUE_ROW /
B.Ele.Th	4	BELIN_AMBROSIO_LOWER_ELEVATION_LABEL_VALUE_ROW /
PPI Min	0.8	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX /
PPI Avg	1.09	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX /
PPI Max	1.5	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX /
ARTmax	397	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX /
Df	0.33	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP /
Db	-0.41	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP /

Canonical field	Value	Provenance
Dp	1.26	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP /
Dt	0.65	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP /
Da	0.83	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP /
Final BAD-D	0.51	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP /
SRAX	13.5	AXIAL_SAGITTAL_CURVATURE_FRONT_GEOMETRIC /

Version provenance

Layer	Version
software	2.0
clinical_policy	CER-AI-2026-09-22-PS3-SOURCE-EXCEPTIONS-V11
source_registry	2026-09-10-ML7-REUSE-CANONICAL-K
srax_algorithm	FRONT_MAP_SRAX_GT20-2026-09-07

OS — PASS

Randleman / ERSS

Component	Points / disposition	Exact finding
Topography	0	
Residual stromal bed	0	
Age	0	
Preoperative pachymetry	0	
Manifest MRSE	0	
Total	0	
Topography category	NORMAL_SYMMETRIC	
Disposition	PASS	

NICE

Component	Input	Points
K2	42.1	1
central_pachymetry	519	2
B_Ele_Th	4	1
I_S	-0.04	1
Total		5
Classification	CAUTION	CAUTION
Decision trigger	CAUTION	NICE total: 5

PS3

Factor	Status	Exact finding
anterior_km	NORMAL	Anterior Km 41.4 D < 48 D.
thinnest	NORMAL	Thinnest 517 µm >=500 µm.
elevation	NORMAL	F.Ele.Th 2 µm <=12 and B.Ele.Th 4 µm <=15.
inter_eye_asymmetry	NORMAL	Inter-eye score 1/5; exceeded: anterior Km.
srax	NORMAL	Front-map SRAX 18.9°; signed I-S -0.04 D is negative (superior asymmetry), so SRAX does not create an inferior-steepening PS3 factor.
ppi_average	NORMAL	PPI Average 1.06 <= 1.20.
corneal_thickness_map_morphology	NOT_EVALUATED	Corneal Thickness Map morphology: not evaluated; surgeon review required; not counted in automated PS3.
relative_thickness_map	NOT_EVALUATED	Relative Thickness Map: not evaluated; surgeon review required; not counted in automated PS3.
pti_ctsp_morphology	NOT_EVALUATED	PTI/CTSP thickness-profile morphology: not evaluated; surgeon review required; not counted in automated PS3.
Moderate / High	0	0
Procedure disposition	PASS	Raw PS3 PRK disposition: ALLOWED.

Astigmatic disparity validation

Status	Magnitude difference	Axis difference
NORMAL	0.8	6
Interpretation	Astigmatic disparity within validation thresholds: magnitude difference 0.80 D; axis difference 6.0°; no PS3 consequence.	Non-scoring; no independent procedure restriction

Belin/Ambrósio BAD-D

Parameter	Value	Interpretation
Final BAD-D	0.83	NORMAL / PASS
Df	0.06	NORMAL / < 1.60; information only
Db	-0.63	NORMAL / < 1.60; information only
Dp	1.05	NORMAL / < 1.60; information only
Dt	0.63	NORMAL / < 1.60; information only
Da	1.14	NORMAL / < 1.60; information only
PPI Min	0.71	NORMAL / < 0.80; information only
PPI Avg	1.06	NORMAL / < 1.08; information only
PPI Max	1.42	SUSPICIOUS / 1.40-1.52; information only
ARTmax	363	SUSPICIOUS / 357-368 um; information only

Procedural safety

Parameter	Canonical result	Exact finding
LASIK_RSB_um	Not documented	
PRK_RST_um	422	
LASIK_PTA_percent	Not documented	
PRK_PTA_percent	18.375	
estimated_final_Kmean_D	39	
scalar_final_Kmean_mode l_valid	Yes	
final_Kmean_model	MYOPIC_0.8_D_PER_ D_ESTIMATE	
hyperopic_review_level	Not documented	
hyperopic_treatment_com ponent_D	-2.75	
hard_stops	preop_thickness: No; sphere_magnitude: No; lasik_rsb: No; lasik_pta: No; prk_rst: No; prk_pta: No; final_kmean: No	
missing	Not documented	
status	PASS	

Procedure planning

Planning item	Canonical result
OS — PASS — PRK	OZ 6.5 mm — TZ 9 mm — Ablation 45 µm
Selected LASIK plan	Not documented
sequence	Not documented
rejection_reasons	Not documented
MMC guidance	RECOMMENDED

Decision basis

Level	Canonical decision driver
CAUTION	key: nice; status: CAUTION; detail: NICE total: 5

Canonical Pentacam values and provenance

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K1	40.8	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT /
K2	42.1	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT /
K2 axis	84	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT /
Km	41.4	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT /
Astigmatism	1.3	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT /
Displayed steep/astigmatic axis	84	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT /
PS3 BAD Axis (flat meridian, beside K1)	174	BELIN_AMBROSIO_CENTER_NUMERIC_BOX /
ISV	15	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM /
IVA	0.09	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM /
KI	1	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM /
CKI	1	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM /
IHA	1.6	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM /
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Canonical field	Value	Provenance
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Dt	0.63	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP /
Da	1.14	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP /
Final BAD-D	0.83	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP /
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Version provenance

Layer	Version
software	2.0
clinical_policy	CER-AI-2026-09-22-PS3-SOURCE-EXCEPTIONS-V11
source_registry	2026-09-10-ML7-REUSE-CANONICAL-K
srax_algorithm	FRONT_MAP_SRAX_GT20-2026-09-07